

8.6

a.

$$H_0: \sigma_M^2 = \sigma_F^2$$

$$H_a: \sigma_M^2 \neq \sigma_F^2$$

$$\alpha = 0.05$$

$$\text{test statistic: } F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \stackrel{H_0}{\sim} F(593, 419)$$

$$RR = \{ F > 1.1968 \text{ or } F < 0.8391 \}$$

$$F^* = \frac{\frac{99.161}{593}}{12.024^2} = 1.19 \notin RR, \text{ do not reject } H_0$$

b.

$$H_0: \sigma_M^2 = \sigma_J^2$$

$$H_a: \sigma_M^2 \neq \sigma_J^2$$

$$\alpha = 0.05$$

$$\text{test statistic: } F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_J^2} \stackrel{H_0}{\sim} F(595, 395)$$

$$RR = \{ F \mid F > 1.1954 \text{ or } F < 0.8394 \}$$

$$F^* = 1.19 \notin RR, \text{ do not reject } H_0$$

c.

$$\text{test statistic } NR^2 \sim \chi^2(4)$$

$$RR = \{ NR \mid NR > \chi_{0.95}^2(4) = 9.488 \}$$

$$NR^* = 59.03 \in RR, \text{ reject } H_0, \text{ 臭臭的臭臭}$$

d.

$$NR^{2*} = 9.82 \in RR, \text{ 臭臭}$$

e.

EDUC, intercept 中 $\text{robse} > \text{se} + \text{CI}$ 臭臭

EXPER, EXPER, FEMALE 中, $\text{robse} < \text{se} + \text{CI}$ 臭

因 robse 比 se 更準確, 且可能大於 $\text{se} + \text{CI}$ 沒有不一致

f. 沒有衝突

(b) MARRIED = 1 or 0 已婚 vs. 未婚, se 可能不同 (臭臭)

(1) 已婚是否 WAGE 不影響

8.1b

a.

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
    Min       1Q   Median       3Q      Max
-1198.14  -295.31   17.98   287.54  1549.41

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    169.775   -2.306  0.0221 *
income       14.201      1.800     7.889 2.10e-13 ***
age          15.741      3.757     4.189 4.23e-05 ***
kids        -81.826     27.130    -3.016  0.0029 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,    Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

2.5 % 97.5 %
kids -135.3298 -28.32302

--- 解讀 ---
KIDS 的係數估計值: -81.8264
95% 信賴區間: [-135.3298, -28.3230]
在其他變數不變的情況下，每多一個小孩對里程數的影響。

d.

```
t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.548    142.6548 -2.7447 0.0066190 **
income       14.2013     1.9389   7.3246 6.083e-12 ***
age          15.7409     3.9657   3.9692 0.0001011 ***
kids        -81.8264     29.1544 -2.8067 0.0055112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

KIDS 係數: -81.8264
穩健標準誤: 29.1544
95% 穩健信賴區間: [-139.3230, -24.3299]

(a) OLS 信賴區間: [-135.3298, -28.3230], 寬度 = 107.0068
(d) 穩健信賴區間: [-139.3230, -24.3299], 寬度 = 114.9931
```

e.

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation_gls,
    weights = 1/(income^2))

Weighted Residuals:
            1Q      3Q      Max
    2488    4.3832   18.5462

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  121.444    -3.500  0.000577 ***
income       1.481     9.420 < 2e-16 ***
age          3.025     5.527  1.03e-07 ***
kids        -76.806     21.848  -3.515 0.000545 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

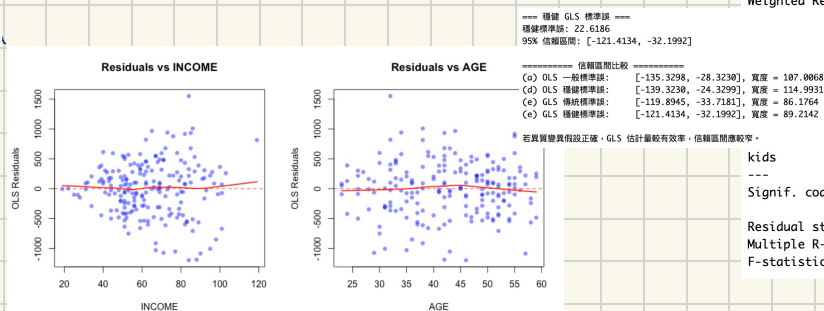
Residual standard error: 6.765 on 196 degrees of freedom
Multiple R-squared:  0.4573,    Adjusted R-squared:  0.449
F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

=== 傳統 GLS 標準誤 ===
KIDS 係數: -76.8063
95% 信賴區間: [-119.8945, -33.7181]

```
t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -424.9962    95.8035 -4.4361 1.526e-05 ***
income       13.9473     1.3470 10.3545 < 2.2e-16 ***
age          16.7175     2.7974  5.9761 1.061e-08 ***
kids        -76.8063    22.6186 -3.3957 0.0008286 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b.



c.

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation_sorted[1:n1,
  ])

Residuals:
    Min       1Q   Median       3Q      Max
-684.07  -245.39    8.69   202.87   631.43

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -392.511    214.166   -1.833  0.07030 .
income       10.960      3.770     2.907  0.00464 **
age          18.869      3.783     4.988 3.14e-06 ***
kids        -70.371     29.138    -2.415  0.01785 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 319 on 86 degrees of freedom
Multiple R-squared:  0.309,    Adjusted R-squared:  0.2849
F-statistic: 12.82 on 3 and 86 DF,  p-value: 5.31e-07
```

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation_sorted[(n -
  n2 + 1):n, ])

Residuals:
    Min       1Q   Median       3Q      Max
-1215.44  -426.21    73.56   304.71  1602.70

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -476.803    548.833   -0.869  0.3874
income       15.556      5.450     2.855  0.0054 **
age          16.388      7.385     2.219  0.0291 *
kids        -116.017     49.861    -2.327  0.0223 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 562 on 86 degrees of freedom
Multiple R-squared: 0.1514, Adjusted R-squared: 0.1218
F-statistic: 5.116 on 3 and 86 DF, p-value: 0.002642

SSE (低所得組): 8750039.6220, df = 86
SSE (高所得組): 27160653.6076, df = 86

GQ 統計量 = 3.1041
F 臨界值 (alpha=0.05, df1=86, df2=86) = 1.4286
p-value = 0.000000
結論: 拒絕 H0，有顯著證據顯示存在異質變異。

